WaferDX: An Explainable AI Decision Support System for Semiconductor Wafer Defect Diagnosis and Process Root Cause Analysis

Abhirup Sain (T14H06318) · David Spickenheuer (T14H06329) · Martin Werner (T14H06319) · Matti Lehmann (T14H06309) · Staniya Thomas (T14H06310)

National Taiwan University — Data Analysis and Machine Learning with Python, Spring 2026

Abstract

In semiconductor manufacturing, a silicon wafer passes through 400–600 process steps before electrical testing produces a spatial map of passing and failing dies. The pattern of those failures encodes the identity of the process tool that caused them: a center-cluster pattern implicates CVD deposition drift; a linear scratch track points to robotic handling damage. Manually interpreting these maps is both error-prone and a bottleneck in yield recovery. Existing automated systems either flag anomalies without classifying them, or classify without explaining which spatial region drove the decision — leaving engineers no better informed about root cause.

WaferDX is an explainable AI decision support system that accepts a wafer map image, classifies it against eight defect pattern types from the WM-811K dataset (811,457 maps, 46,393 lots), and pairs each prediction with a Grad-CAM spatial attribution heatmap and a knowledge-based root cause hypothesis. The system is implemented in Python using TensorFlow/Keras for the CNN backbone and Streamlit for the interactive interface, allowing engineers to upload a wafer image or draw one directly on a canvas.

Three design trade-offs shaped the final architecture. First, a direct CNN classifier (96×96 input) was retained over a segmentation-then-classify pipeline after comparative evaluation showed the pipeline collapsing to a single dominant class prediction on the 160-sample balanced test set, achieving only 11.9% accuracy. Second, data augmentation via rotation and flipping was evaluated as a separate model variant; it improved Edge-Loc accuracy from 30% to 85% but destroyed Donut accuracy, dropping it from 65% to 10%, because rotation augmentation removes the orientation cues that distinguish radially symmetric defect types. Third, a knowledge-based root cause mapping was used instead of a learned one, since no public dataset links WM-811K labels to confirmed fab process failures.

Evaluation tested four model variants on 160 generated wafer images — 20 per class across Center, Donut, Edge-Loc, Edge-Ring, Loc, Near-full, Random, and Scratch. The Fullstack model achieved the highest overall accuracy at 17.5%, while the Best CNN reached 13.1% and the augmented variant 13.8%. Per-class results reveal severe calibration failures: the Best CNN scores 65% on Donut but 0% on Center, Edge-Ring, Loc, and Scratch. These numbers drove the system's core advisory design: all outputs display the full probability distribution across all eight classes, the Grad-CAM heatmap for visual verification, and an explicit multi-hypothesis panel when the confidence gap between top-1 and top-2 falls below 0.15.

I. Application Context and Decision Problem

I.I. System Application Scenario

In a 300mm semiconductor fab, a single production lot contains 25 wafers. After front-end electrical testing, each wafer produces a binary die map: pass (1) or fail (2) at every die location on the circular substrate. The spatial distribution of failures — not their total count — is what matters for process diagnosis. A "Center" pattern (a dense cluster of failing dies in the wafer center) implicates CVD deposition rate non-uniformity at the center of the reactor chamber. A "Scratch" pattern (a linear diagonal track of failing dies) points to mechanical contact during wafer handling between process steps, most commonly at the robot arm end-effector or inside a wafer boat. Identifying the correct pattern quickly and correctly determines how fast a yield engineer can isolate the faulty tool and prevent additional wafers from being processed under the same failure condition.

I.II. Decision-Making Challenges

Engineers face three compounding difficulties. First, several defect patterns are visually ambiguous: Donut (a hollow annular ring of failures) and Edge-Ring (a continuous ring of failures concentrated at the wafer perimeter) both produce annular distributions but implicate different process steps — CMP pressure ring effect versus RTP thermal non-uniformity, respectively. Second, real production wafers can exhibit multiple overlapping defect patterns simultaneously, a scenario the WM-811K dataset labels as single-class but which engineers encounter routinely. Third, when a classification system outputs only a label without a spatial explanation of where it found the evidence, engineers have no way to verify whether the result is trustworthy or driven by image artifacts at the wafer edge.

I.III. Limitations of Existing Approaches

Current fab-floor inspection tools fall into two categories. Automated Optical Inspection (AOI) systems like KLA's Klarity detect individual die failures and flag their locations, but do not classify the aggregate spatial pattern and provide no process hypothesis. Academic models trained on WM-811K — such as the ResNet-50 ensemble reported by Shim et al. (2020) achieving 98.7% on the standard test split — optimize for benchmark accuracy on a class-balanced subset that does not reflect production class distributions, where fewer than 300 Scratch samples exist in the full 811K labeled set. Neither type of system provides the spatial explanation that would let an engineer validate a classification decision before acting on it.

I.IV. What WaferDX Supports

WaferDX supports one primary decision: *which process tool or step is most likely responsible for the observed die failure pattern on this wafer?* The system does not make this call automatically. It outputs a ranked probability distribution over all eight defect classes, a Grad-CAM heatmap showing which spatial regions of the wafer most influenced the top prediction, a root cause mapping from defect class to process step, and — when the confidence gap between the top-1 and top-2 predictions is less than 0.15 — an explicit dual-hypothesis display that tells the engineer "this map is ambiguous between Edge-Ring and Donut; here is what each implies for your process." The engineer decides what to do next.

II. Related Design Approaches

II.I. Single-Label End-to-End CNN Classifiers

The dominant published approach trains a convolutional neural network end-to-end on WM-811K with a softmax output layer across nine classes (eight defect types plus “None”). Published accuracy numbers are high: Wu et al. (2015) achieved 92.4% using a combined hand-crafted feature and k-nearest-neighbor approach, and Shim et al. (2020) reported 98.7% using an ensemble of three CNNs on the standard test split. The critical limitation is not the model architecture but the evaluation setup. The standard WM-811K split samples proportionally from the original distribution, in which 47.8% of all labeled wafers belong to the “None” class and “Scratch” accounts for only 1.1% of labeled samples. A model that predicts “None” for every ambiguous input can appear highly accurate while providing zero utility to an engineer who needs to distinguish between Scratch and Edge-Loc. None of these published systems generate spatial explanations or confidence-aware multi-hypothesis outputs.

II.II. Segmentation-Then-Classify Pipelines

A more interpretable design separates detection from classification: a segmentation network first produces per-pixel activation maps for each defect class, and a second classifier operates on those maps rather than on the raw wafer image. The appeal is that the intermediate segmentation output is inherently spatial and interpretable — engineers can see which pixel regions activate for each class hypothesis. This architecture (labeled “Seg $\to$ CLF” in our evaluation) was implemented and evaluated alongside the direct CNN. The practical result was 11.9% overall accuracy on the 160-sample balanced test set, compared to 13.1% for the direct CNN without augmentation. More diagnostically, the Seg $\to$ CLF pipeline predicted “Center” with confidence 1.0 for over 90% of all inputs regardless of true class, indicating that the classification head overfit to the dominant class in its training distribution rather than learning from the segmentation activation features.

II.III. Where WaferDX Differs

WaferDX does not treat accuracy as the only criterion for system usefulness. A model that achieves 65% on Donut but 0% on Center, Edge-Ring, Loc, and Scratch is not appropriate for single-hypothesis output — and the system makes this explicit. Rather than presenting a single classification label, the UI always shows the full 8-class probability distribution and flags outputs where model behavior is known to be unreliable. The Grad-CAM spatial attribution layer provides the visual explanation that benchmark-focused systems omit entirely, allowing engineers to sanity-check whether the model is attending to the actual defect region or to irrelevant image structure at the wafer boundary.

III. Artifact Overview

III.I. Overall System Functions

WaferDX performs four distinct functions:

1. **Input handling** — accepts a wafer map as either an uploaded PNG file or as a user-drawn image on an interactive canvas widget (`streamlit-drawable-canvas`), supporting both raw fab export images and synthetic defect sketches for testing.
2. **Classification** — runs the trained CNN (TensorFlow/Keras, 96×96 input, trained on a WM-811K subset) and produces a softmax probability vector across 8 defect classes. Both the Best CNN and the augmented variant are available in the codebase.

3. **Spatial explainability** — computes a Grad-CAM heatmap from the final convolutional layer by computing the gradient of the top predicted class score with respect to layer activations, globally average-pooling to obtain per-channel weights, and linearly combining the resulting activation maps. The heatmap is upsampled to 96×96 and overlaid on the input image as a red-to-blue color map.
4. **Root Cause Advisory** — maps the top predicted defect class to a process-level hypothesis using a hand-curated knowledge base. Output includes the likely process step, the implicated tool type, and a concrete recommended inspection action (e.g., for a Scratch prediction: *"Inspect robot arm end-effector and CMP pad condition"*).

III.II. System Boundaries

WaferDX processes single wafer map images. It does not connect to fab MES systems, lot history databases, or real-time tool sensor feeds. Input images are not stored between sessions. The system does not model lot-to-lot or wafer-to-wafer trends, does not account for grade curves or inspection tool calibration state, and does not produce scrap or rework recommendations. The root cause mappings are knowledge-based heuristics from standard CMOS process engineering literature — they are advisory hypotheses, not confirmed diagnoses. Multi-defect wafer maps (two or more overlapping defect patterns) are a known out-of-scope case; the system outputs the single most-probable class rather than a multi-label prediction.

III.III. Usage Workflow

[1] **User uploads PNG or draws wafer on canvas**
↓
[2] **Preprocessing pipeline**
grayscale → zero-pad to square → nearest-neighbor resize to 96×96
normalize: `round(arr / 127.5) / 2.0` → values ∈ {0.0, 0.5, 1.0}
↓
[3] **CNN forward pass** → softmax over 8 defect classes
↓
[4] **Grad-CAM** — gradient of top class score w.r.t. final conv layer activations
→ weighted activation sum → upsample to 96×96 → overlay heatmap
↓
[5] **Display** — ranked probability bars + heatmap overlay on input image
↓
[6] **RCA lookup** — defect class → process hypothesis + recommended action
↓
[7] **Multi-hypothesis view** if top-1 confidence gap < 0.15 or confidence < 0.5
↓
[8] **Engineer interprets and decides**

Listing 1: WaferDX processing pipeline from wafer map input to root cause advisory output. Steps 1–7 are fully automated; step 8 is human-only.

The preprocessing step deserves attention. WM-811K wafer maps span 632 distinct spatial formats, ranging from 25×25 to 300×300 pixels with varying aspect ratios. Simple bilinear rescaling averages neighboring pixels and corrupts the discrete 3-value integer encoding (0 = background, 1 = passing die, 2 = failing die). WaferDX applies nearest-neighbor resampling after zero-padding to a square aspect ratio, preserving the integer encoding throughout. Input size of 96×96 was chosen above the dataset's average map size to retain spatial detail on larger wafers without making the network unnecessarily deep.

IV. Design Decisions and Trade-offs

IV.I. Decision 1: Direct CNN vs. Segmentation-Then-Classify Pipeline

The segmentation-then-classify (Seg → CLF) architecture was the theoretically preferred design at the project's outset. A segmentation network producing per-pixel activation maps for each class would make the model's spatial reasoning directly visible, eliminating the need for post-hoc attribution methods like Grad-CAM. In practice, the Seg → CLF pipeline achieved 11.9% overall accuracy on the 160-sample balanced test set — lower than the direct CNN at 13.1% and substantially below the Fullstack model at 17.5%. The failure mode was qualitative: inspection of the evaluation CSV showed the classification head predicting "Center" with confidence 1.0 for the majority of inputs including Scratch, Random, and Edge-Ring wafers. The segmentation activations were not providing discriminative signal to the downstream classifier, most likely because the two network components were trained independently without end-to-end gradient flow.

Why the direct CNN was chosen: Grad-CAM produces a spatial attribution map that, while post-hoc, is mathematically grounded and visually interpretable for engineers who understand wafer spatial structure. The direct CNN also trains as a single optimization problem, making debugging simpler and training more stable than a disjoint two-stage pipeline.

Rejected alternative: End-to-end joint training of segmentation and classification heads with shared gradients would address the pipeline's failure mode. This was scoped out due to the additional architecture redesign and compute time required within the project timeline.

Conditions and limits: For genuinely multi-defect wafers — two or more distinct spatial patterns coexisting on one map — the direct CNN with Grad-CAM produces a single attribution map that conflates both patterns. A segmentation architecture trained end-to-end remains the architecturally correct choice for the multi-label production scenario.

IV.II. Decision 2: Class-Agnostic vs. Class-Selective Data Augmentation

Rotation and flipping are standard augmentation techniques for image classification, and applying them uniformly appeared safe. The evaluation result was sharply non-uniform. The augmented model (Best CNN + Aug) improved Edge-Loc accuracy from 30% to 85% — a class where the localized edge failure can appear at any angular position around the wafer perimeter, so orientation invariance genuinely helps. But Donut accuracy collapsed from 65% to 10%. Donut is a radially symmetric annular ring, and its class identity relies on the ring being complete. Rotating a partial Donut sample during training teaches the model that a partial ring is a valid Donut — which conflicts with the actual defect definition and confuses the model on Donut inputs at inference time.

Why augmentation was retained as an option: The augmented model is available in the codebase and UI alongside the non-augmented variant, allowing engineers or future developers to select the model appropriate for their dominant defect class distribution.

Rejected approach: Uniform augmentation across all 8 classes. The correct approach, identified after evaluation, is class-selective augmentation: apply rotation/flip only to orientation-insensitive classes (Edge-Loc, Scratch, Random, Loc) and exclude radially symmetric classes (Donut, Edge-Ring, Center, Near-full). This was not implemented within the project timeline.

Conditions and limits: This finding is specific to spatially structured classification problems where class geometry determines valid augmentation operations. In natural image recognition, uniform rotation augmentation reliably improves generalization; in wafer map classification, it actively damages performance on specific classes.

IV.III. Decision 3: Knowledge-Based Root Cause Mapping vs. Learned Mapping

Linking defect pattern classification to specific process root causes requires labeled examples of confirmed causal chains: *this wafer showing an Edge-Ring pattern was caused by this RTP chamber running at the wrong temperature ramp rate.* No such public dataset exists. The SECOM dataset from UCI (1,567 samples, 590 process features) contains semiconductor manufacturing sensor data but does not include spatial wafer maps or defect class labels. Constructing a learned mapping was therefore not feasible without proprietary fab data.

Why knowledge-based mapping was chosen: The mapping from defect class to process hypothesis is well-established in semiconductor process engineering literature and does not require training data. Each defect class maps to one or two candidate process failures:

Defect Class	**Process Hypothesis**
Center	CVD deposition rate drift at wafer center; spin-coat nozzle blockage
Donut	Non-uniform spin coating (radial thickness variation); CMP pressure ring effect
Edge-Ring	Non-uniform thermal processing (RTP); irregular photoresist coating at wafer edge
Edge-Loc	Localized thermal non-uniformity (RTP furnace); local gas flow asymmetry
Loc	Local particle contamination; clogged process nozzle
Random	Airborne contamination; clean room excursion event
Scratch	Mechanical handling damage (robot end-effector); CMP over-polishing
Near-full	Catastrophic process excursion; full process step failure

Table 1: Knowledge-based root cause mapping. Each defect class links to one or two candidate process failure mechanisms from standard CMOS front-end-of-line process engineering literature.

Rejected alternative: Using a large language model at inference time to generate root cause suggestions was discussed and rejected. LLM outputs for process engineering recommendations are plausible-sounding but unverifiable without ground-truth validation, and presenting them as system output would create a misleading impression of diagnostic authority.

Conditions and limits: The knowledge-based mappings are calibrated for standard bulk CMOS front-end processes. They may not apply to III-V compound semiconductor processes, MEMS fabrication, or advanced packaging, where failure mode mechanisms differ substantially.

V. AI Role and System Autonomy

V.I. The Role of AI in WaferDX

The AI component performs one function: given a 96×96 grayscale wafer map encoded as a $\{0.0, 0.5, 1.0\}$ tensor, produce a probability distribution over eight spatial defect pattern classes. The model is a convolutional neural network trained on a WM-811K subset using TensorFlow/Keras. It operates entirely on spatial patterns in the 2D pixel grid — it knows nothing about process parameters, lot history, tool identifiers, or time. Grad-CAM augments this output with a spatial attribution map by computing the gradient of the top class score with respect to the final convolutional layer's activations, globally average-pooling those gradients to produce per-channel importance weights, and taking a weighted linear combination of the activation maps. The result is a heatmap showing *where* the model is looking, not *why* the defect occurred.

The root cause advisory layer is rule-based, not learned. Once the CNN produces a top predicted class, a Python dictionary lookup maps that class to the corresponding process hypothesis from Table 1. No machine learning is involved in this step.

V.II. Level of Automation

Classification and heatmap generation are fully automatic: no human input is needed between image submission and probability output. The root cause lookup is also automatic. Critically, all three outputs arrive together as a package — the engineer sees the probability distribution, the spatial heatmap, and the process hypothesis simultaneously, not as a sequential decision tree.

The decision layer — whether to initiate a tool inspection, quarantine the lot, escalate to senior engineering review, or flag the result as low-confidence and seek a second opinion — is entirely human. WaferDX never issues a recommendation to act. It provides structured analytical input to support the human decision.

V.III. How Engineers Intervene and Review

Three safeguards are built into the system interface. First, the full 8-class probability distribution is always visible as a ranked bar chart, not just the top-1 label with its confidence score. An engineer can see immediately that a "Donut" prediction came in at 57% confidence with Edge-Loc as the second candidate at 38% — a distribution that should prompt caution before initiating CMP maintenance. Second, the Grad-CAM heatmap allows a visual sanity check: if the model is attending to the wafer edge rather than a central cluster when predicting "Center," the engineer can identify the misattribution directly. Third, the system automatically triggers a dual-hypothesis display when the confidence gap between top-1 and top-2 falls below 0.15, presenting both process hypotheses side-by-side with their respective confidence scores. All output text in the UI labels results as "pattern hypothesis" rather than "diagnosis" or "root cause," making the advisory scope explicit throughout.

VI. Evaluation System, Logic and Evidence

VI.I. Goals and Evaluation Setup

The evaluation has two goals: (1) compare classification accuracy across four model variants to determine which is most suitable for deployment, and (2) characterize failure modes to identify where the multi-hypothesis display threshold and confidence warnings should be applied. The test set consists of 160 PNG images generated from WM-811K using `generate_images.py`, balanced at 20 samples per class across all 8 defect types. Four models were evaluated using `test.py`, which loads each model, preprocesses each image to the model's required input size and normalization scheme, runs inference, and records the full softmax probability vector, top-3 predicted labels, and correctness flag for each (image, model) pair into a CSV file (`evaluation_results.csv`).

The evaluation baseline is random chance: for an 8-class balanced problem, random prediction yields 12.5% accuracy. Any model performing near or below this level on the balanced test set has not learned generalizable spatial features for that class.

VI.II. Per-Class and Overall Accuracy

Class	Best CNN	Best CNN + Aug	Fullstack	Seg → CLF
Center	0%	0%	—	—
Donut	65%	10%	—	—
Edge-Loc	30%	85%	—	—
Edge-Ring	0%	0%	—	—
Loc	0%	0%	—	—
Near-full	5%	15%	—	—
Random	5%	0%	—	—
Scratch	0%	0%	—	—
Overall	**13.1%** (21/160)	**13.8%** (22/160)	**17.5%** (28/160)	**11.9%** (19/160)

Table 2: Per-class and overall accuracy for all four model variants on the 160-sample balanced test set (20 images per class). Fullstack and Seg → CLF per-class breakdowns are not shown due to space, but overall figures are confirmed from evaluation_results.csv. "—" indicates data not broken out for this model in the per-class analysis.

The overall figures cluster between 11.9% and 17.5% — near random-chance on a balanced 8-class problem. This reflects a genuine model performance issue, not a test set artifact: the same CNN architecture that reportedly achieves 70–80% accuracy during training on the full WM-811K training set generalizes poorly to this balanced 160-sample test set, where rare classes like Scratch and Random are equally represented with common ones like Donut and Edge-Loc.

VI.III. Key Failure Cases

The Center/Near-full confusion is the most operationally consequential failure. The Best CNN assigns Near-full probability 0.90–0.98 to Center inputs: 18 out of 20 Center images are predicted as Near-full with high confidence. The root cause implications are entirely opposite — Center implicates a routine CVD deposition calibration drift, while Near-full implicates a catastrophic full-step failure requiring immediate lot quarantine. A system presenting only top-1 outputs would mislead an engineer into a drastic response for a routine calibration issue on 90% of Center wafers.

The augmentation experiment's Donut collapse is the second major failure. Edge-Loc accuracy jumps from 30% (no augmentation) to 85% (with augmentation), but Donut drops from 65% to 10% in the same model. The augmented model was not evaluated on Donut-heavy production input distributions because the effect was discovered only after examining per-class results in the CSV.

The Seg → CLF pipeline's collapse to "Center" predictions (confidence = 1.0 for over 90% of inputs) is a training failure visible in every row of the evaluation CSV for that model variant. The classification head learned the dominant class from its training distribution rather than exploiting the segmentation activations.

VI.IV. Evaluation Limitations

The 160-sample test set is balanced by design, which does not reflect the actual production input distribution, where Center and Near-full patterns are rare events and "None" (no defect) is by far the most common outcome. Model performance on rare classes in a production setting would be expected to be lower still, since training data for Scratch (fewer than 300 samples in the full labeled WM-811K dataset) is genuinely scarce. All test images are single-class; no multi-defect synthetic maps were included in the evaluation, so the system has not been tested on the scenario it would encounter most in a real fab. The evaluation is also entirely automated — no engineer reviewed the Grad-CAM outputs qualitatively, so heatmap quality remains unvalidated against expert judgment.

VII. Discussion: Transferable Design Insights

VII.I. Transferable Design Principles

Augmentation must respect class geometry. Rotation augmentation improved Edge-Loc accuracy by 55 percentage points and degraded Donut accuracy by 55 points in the same training run. Any domain where class identity is geometrically defined — satellite image analysis, histopathology slide orientation, printed circuit board defect inspection, medical imaging — requires class-specific augmentation policies. Uniform augmentation is not a safe default when different classes have different spatial symmetries.

Confidence calibration matters more than accuracy for decision support. The Best CNN assigns Near-full confidence of 0.90–0.98 to Center inputs and is wrong on 18 of 20 such cases. That combination — high confidence, wrong answer — is more dangerous for decision support than low confidence with a wrong answer, because the UI's low-confidence warning would not trigger. Post-hoc calibration methods (Platt scaling, temperature scaling, isotonic regression on a held-out calibration set) should be treated as mandatory components in any decision support system, not optional improvements.

Separating classification from root cause reasoning makes systems easier to maintain. The CNN does not need to learn process engineering. The RCA mapping does not need to encode visual feature detection. When a new process step introduces a new failure mode, only the lookup table requires updating, not a model retraining cycle. This modularity pattern applies to any ML system where domain knowledge is well-structured but training data for that knowledge layer is unavailable.

VII.II. Context-Dependent Elements

The preprocessing pipeline — nearest-neighbor resampling, zero-padding before resize, {0.0, 0.5, 1.0} normalization — is specific to WM-811K's integer-encoded format. Applying it to continuous-valued images (standard photographs, analog sensor readouts) would produce incorrect model inputs. The RCA knowledge base is written for standard bulk CMOS front-end-of-line processes. A fab running SiC power devices, GaN-on-Si, or advanced 3D packaging would need entirely different process-to-defect mappings.

The 96×96 input resolution was chosen for WM-811K's size distribution (25×25 to 300×300, mean near 80×80). A dataset with larger wafer maps — as would come from higher-resolution inspection equipment — would require a larger input resolution and a deeper network.

VII.III. Design Risks and Governance Considerations

The most significant governance risk in this system is calibration failure producing high-confidence wrong predictions without triggering any warning. The current threshold-based safeguard (show multi-hypothesis display when confidence gap < 0.15) does not help when the model is confidently wrong about the top-1 class, as observed in the Center/Near-full case. Any production deployment should include: (1) per-class confidence calibration validated on a representative held-out set, (2) mandatory human review for Near-full and Scratch predictions given their high operational consequence, and (3) a logged feedback loop so engineer override decisions accumulate as training data for future model versions.

A secondary risk is over-reliance on the root cause advisory output. The knowledge-based RCA mappings are heuristics, not verified causal chains. An engineer who acts on the process hypothesis without cross-checking with tool run logs and chamber sensor data could initiate a costly tool inspection based on a misclassified defect type. The system's labeling of all RCA outputs as "hypotheses" rather than "diagnoses" is a necessary but not fully sufficient safeguard.

VIII. Conclusion and Future Extensions

VIII.I. System Design Contributions

WaferDX demonstrates that explainability in a wafer defect decision support system requires more than appending a Grad-CAM heatmap to a CNN. The evaluation across four model variants on a 160-sample balanced test set revealed that overall accuracy alone conceals class-level calibration failures severe enough to make single-hypothesis outputs actively misleading — most critically, the Best CNN predicts Near-full with 0.90–0.98 confidence for Center inputs, which would trigger a lot quarantine response for a routine calibration drift.

The system's practical contribution is the decision support framing: always displaying the full 8-class probability distribution, pairing every prediction with a spatial Grad-CAM heatmap that engineers can use to verify model attention, providing a structured RCA advisory layer that converts classification outputs into actionable process hypotheses, and triggering explicit dual-hypothesis display for ambiguous predictions. The finding that class-selective augmentation is required for orientation-asymmetric defect patterns is a concrete, falsifiable result with direct implications for any spatial classification task where class identity is geometrically constrained.

VIII.II. Design Adjustments with Additional Resources

With more time and compute, three extensions would most substantially improve the system.

First, per-class temperature scaling applied to the model's softmax output using a held-out calibration set would address the systematic overconfidence in Near-full predictions. A 500-sample per-class calibration set (4,000 samples total) drawn from WM-811K and processed through isotonic regression would likely reduce the Center/Near-full confidence gap to a level where the existing threshold-based warning system triggers correctly.

Second, synthetic multi-defect map generation — producing maps by taking the logical union of two single-class WM-811K samples — would allow training and evaluation in the multi-label setting that better reflects production reality. This requires a multi-label loss function (binary cross-entropy per class rather than softmax cross-entropy) and multi-label evaluation metrics (per-class recall, F1 per class), but the data generation step is straightforward from the existing dataset.

Third, integrating process log data from confirmed-defect wafers — tool IDs, chamber sensor readings, and run timestamps — would allow the knowledge-based RCA layer to be partially replaced by a learned mapping, potentially validating and extending the current heuristic associations with empirical causal data from actual production. This data exists inside every semiconductor fab but is not publicly available.

IX. References

Chollet, F. (2021). *Deep learning with Python* (2nd ed.). Manning Publications.

Molnar, C. (2022). *Interpretable machine learning: A guide for making black box models explainable* (2nd ed.). https://christophm.github.io/interpretable-ml-book/

Selvaraju, R. R., Cogswell, M., Das, A., Vedantam, R., Parikh, D., & Batra, D. (2017). Grad-CAM: Visual explanations from deep networks via gradient-based localization. *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*, 618–626.

Shim, J., Hwang, C., & Lee, S. (2020). Wafer map defect pattern classification based on convolutional neural network features and error-correcting output codes. *Journal of Intelligent Manufacturing*, *31*(5), 1861–1875.

Wu, M. J., Jang, J. S. R., & Chen, J. L. (2015). Wafer map failure pattern recognition and similarity ranking for large-scale data sets. *IEEE Transactions on Semiconductor Manufacturing*, *28*(1), 1–12.

Gregor, S., & Hevner, A. R. (2013). Positioning and presenting design science research for maximum impact. *MIS Quarterly*, *37*(2), 337–355.